

Figure 5: Mixed plots with semi log axis

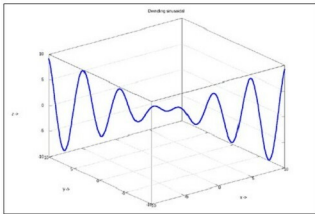


Figure 6: 3-D plot of a dwindling sinusoidal

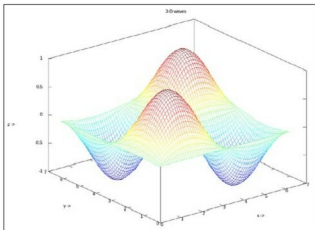


Figure 7: 3-D waves

There are many other ways of drawing interesting 2-D figures for all kinds of mathematical and scientific requirements. So, before closing on 2-D plotting, let's look into just one more often-needed drawing—plotting with the log axis, and also with two y-axes on a single plot. The function for that is *plotyy()*. Note that *plotyy()* is calling the corresponding function pointers *@plot* and

@semilogy passed to it, in the following code segment:

```
octave:1> x = 0:0.1:2*pi;
octave:2> y1 = sin(x);
octave:3> y2 = exp(exp(x));
octave:4> ax = plotyy(x, y1, x, y2, @plot, @semilogy);
octave:5> xlabel("x ->");
octave:6> ylabel(ax(1), "sine ->");
octave:7> ylabel(ax(2), "e^(e^x) ->");
octave:8> title("Mixed plots");
octave:9>
```

3D visualisation

And finally, let's do some 3-D plotting. *plot3()* is the simplest Octave function to use for a simple 3-D drawing, taking the set of (x, y, z) points. Here's the sample code to draw the dwindling sinusoidal curve shown in Figure 6:

```
octave:1> x = -10:0.1:10;
octave:2> y = 10:-0.1:-10;
octave:3> z = x .* sin(x - y);
octave:4> plot3(x, y, z, "b", "LineWidth", 4);
octave:5> xlabel("x ->");
octave:6> ylabel("y ->");
octave:7> zlabel("z ->");
octave:8> title("Dwindling sinusoidal");
octave:9> grid("on");
octave:10>
```

In case you want to plot the values of the 2-D matrix against its indices (x, y), it could be done using *mesh()*, one of the many other 3-D plotting functions.

Figure 7 shows the same, created using the following code:

```
octave:1> x = 0:0.1:2*pi;
octave:2> y = 0:0.1:2*pi;
octave:3> z = sin(x)' * sin(y);
octave:4> mesh(x, y, z);
octave:5> xlabel("x ->");
octave:6> ylabel("y ->");
octave:7> zlabel("z ->");
octave:8> title("3-D waves");
octave:9>
```

What's next?

Next, we will look into Octave from a statistician's perspective. 

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The author is an open source hardware and software hobbyist, with a passion for mathematics. His exploration with mathematics, in every aspect of life, dates back to the 1990s. A gold medalist from the Indian Institute of Science, mathematics and knowledge sharing are two of his many passions. Apart from that, he experiments with Linux and embedded systems to share his learning through his weekend workshops. Learn more about him and his experiments at <http://sysplay.in>. He can be reached at email@sarka-pugs.com.